

Math 1452 Final Exam Fall 2014

*Calculators are not allowed on this exam. Work all questions completely. Show all work as described in class.
Copyright 2014 Dept of Mathematics and Statistics, Texas Tech University. Unauthorized reproduction prohibited.*

1. Consider the region bounded by $y = 2x$, $y = 1$, and the y -axis. **Set up** (but do not solve) integrals to find
 - (a) The volume of the solid generated by rotating this region about the horizontal line $y = -3$ using both shells and washers.
 - (b) The volume of a solid whose base is this region and has the property that each cross section perpendicular to the x -axis is a square.
2. Graph the rose $r = 3 \sin(2\theta)$ and **set up** an integral to find the area enclosed by one leaf.
3. A particle moves with speed $f(t) = 3t^2$. Explain in complete detail why the formula **Distance = Speed * Time** is inadequate to finding the distance traveled by the particle from time $t = 0$ to $t = 4$ and why an integral can be used to find this distance. **Set up** such an integral.
4. Evaluate the following integrals.
 - (a) $\int \frac{1}{x\sqrt{\ln x}} dx$
 - (b) $\int e^{3x} \cos(2x) dx$
 - (c) $\int \sin^3(x) dx$
 - (d) $\int \frac{6x - 4}{x(x + 4)} dx$
5. Evaluate $\int_2^3 \frac{5}{x - 2} dx$.
6. Indicate if the following series converge or diverge. You must identify all the tests you use and show all the work needed to apply them.
 - (a) $\sum_{k=2}^{\infty} k^{1/k}$
 - (b) $\sum_{k=3}^{\infty} \frac{\ln k}{\sqrt{k}}$
 - (c) $\sum_{k=0}^{\infty} \frac{3^k}{(2k)!}$
 - (d) $\sum_{k=0}^{\infty} \frac{1}{k^2 + 1}$
7. Explain completely the difference between absolute and conditional convergence.
8. If $a_k > 0$ and $\frac{a_{k+1}}{a_k} = \frac{3}{k}$ for all $k > 1$, does $\sum_{k=1}^{\infty} a_k$ converge? Why or why not?
9. Find the first 3 terms of the Taylor series for $f(x) = x \ln(x)$ centered at 2.
10. Let $\mathbf{u} = \langle 1, 4, -3 \rangle$ and $\mathbf{v} = \langle 2, 0, -1 \rangle$.
 - (a) Find a unit vector that points in the same direction as $\mathbf{u} - 2\mathbf{v}$.
 - (b) Are $\mathbf{u}$ and $\mathbf{v}$ orthogonal? Explain.